Enhancing Security and Performance in Software-Defined Networks Using AI-Driven Automation

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2025 12th International Conference on Software Defined Systems (SDS) | 979-8-3315-7258-7/25/$31.00 ©2025 IEEE | DOI: 10.1109/SDS68531.2025.11409794

***Abstract*—This study focused on the concept of incorporating Artificial Intelligence (AI) into Software Defined Networks (SDN) to optimize network security and performance. This paper tackles some of the problems with managing Software Defined Networks by introducing a framework that uses supervised learning and reinforcement learning algorithms for automated AI control. From a host of simulated experiments and actual case analyses conducted by the study, the results indicated that AI automation was able to help the SDN controller actively identify anomalies within the network, optimize routing patterns for better performance, and react to threats automatically and instantly. Specifically, quantitative analysis indicated a positive improvement of 27% accuracy of threat identification, 20% reduction of latency times, and 15.8% boost of throughput while simultaneously registering a 60% reduction of reliance on administrative activities. This study further introduced a unified AI-SDN management system that resolved the long-standing issue of the separation between performance optimization and network security. Through the application of intelligent decision-making algorithms, the system was able to simultaneously guarantee safety and optimize resource utilization regardless of attacking or traffic surge scenarios. Although the study validated the feasibility of AI-powered SDN, it did point out some limitations concerning diversity in datasets, computational complexity, and interpretability. Towards addressing such limitations, the authors suggested incorporating micro-AI models at the edge layer, common datasets, and open governance structures. The study finds that artificial intelligence-powered automation is a revolutionary improvement over SDN and can place this technology at the forefront of future intelligent, self-adaptive, and secure infrastructural architectures for 5G communications, IoT, and cloud services.**

Keywords—Software-Defined Networking (SDN), Artificial Intelligence (AI), Machine Learning, Network Security, Automation, Performance Optimization, Reinforcement Learning, Intrusion Detection, Network Orchestration, Edge Computing.

I. Introduction

A new technological trend known as software-defined networking (SDN) emerged because of this accelerated rate of innovation. SDN offers a paradigm that divides the forwarding-plane operation from the control-plane function, allowing centralized network control. This contrasts with traditional networks, which recognized for being heavily dependent on hardware-centric designs and static settings. The ability of SDN technology centrally manage a network for improved performance and simple traffic pattern adaptation is one of its main advantages for businesses. The dependence on software control planes and APIs creates new problems for network security and performance management.

With the rising trend of cloud computing, edge computing, and Internet of Things (IoT) technology integration into computer networks, the probability of a complex network becoming a target for advanced cyber-attacks continues to increase. Conventional cybersecurity solutions may fail to perform well under these changing network dynamics due to a lack of real-time intelligence to counter new cyber threats. In addition, performance-related issues arising from SDN may degrade the efficiency of services for higher-priority applications. All this indicates that there is a great need for the integration of intelligent techniques into a software-defined network for efficient and secure operation.

AI and Machine Learning (ML) have been recognized as promising solutions to counteract the issues. AI-powered automation presents possibilities for predictive analytics, anomaly detection, and smart decision-making that enable self-optimization of the network and proactive reactions to possible dangers. Analysis of traffic patterns in conjunction with vulnerability analysis and autonomous enforcement of Security Policies by AI-powered SDN can introduce remarkable improvements to network security and performance. AI-powered automation not only lessens reliance on manual settings but also introduces rapid reaction times with lower chances of human error and optimized resource utilization for complex network topologies.

This study investigates the convergence of AI, automation, and Software Defined Networks with a focus on techniques further improve network Security and Performance. The study surveys current AI-powered techniques for securing and managing network performance using AI and analyzes their efficacy on current network settings. This study aims to introduce a method to build a resilient, high-performance, and secure network that satisfies the requirements of the current digital age by marrying the power of AI with the flexibility of Software Defined Networks.

II. Research Problem

Although Software Defined Networks (SDN) are quite flexible and scalable, nowadays network architectures confronted with challenges to assure strong security and good performance at the same time. On one hand, centralized control of the network due to the SDN concept poses some risks for organizations. This can be attributed to the danger of attacking the controller of the network directly or using the API of the network, as well as traffic patterns. On the other hand, congestion, latency, and suboptimal utilization of resources continue to pose a threat to good services.

III. Research Objectives

The main aims of the study are:

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- To analyze the current security and performance challenges in Software-Defined Networks.
- Investigating AI-powered automation methodologies for the improvement of SDN security and performance.
- To evaluate the effectiveness of integrating AI-based predictive analytics, anomaly detection, and intelligent decision-making into SDN environments.
- Propose a framework for incorporating automated optimization techniques for security and performance for SDN architectures.

IV. Research Questions

This research aims to answer the following key questions:

- What are the key security issues and performance problems with existing architectures for SDN?
- In what ways AI and machine learning algorithms be harnessed for automating security and improving performance in an SDN?
- Is there a measurable benefit to incorporating automation with AI technology into network security and performance with SDN?
- Are there best practices that can help with the implementation of AI automated solutions for applications of Software Defined Networks?

V. Literature Review

Software-Defined Networking (SDN) has become a transformative approach in the design and management of modern networks. By separating the control plane from the data plane, SDN allows centralized network control, programmability, and dynamic management of resources [1]. This paradigm shift contrasts sharply with traditional networking, which relies on static configurations and hardware-based switches and routers. SDN enables real-time network adjustments, simplified policy enforcement, and rapid deployment of new applications, making it a cornerstone technology for cloud computing, edge computing, and large-scale enterprise networks [2].

Despite its advantages, SDN introduces new security vulnerabilities and performance challenges. The centralized controller, a key component of SDN, represents a single point of failure that can be exploited by attackers [3]. Security threats in SDN include Distributed Denial of Service (DDoS) attacks targeting the controller, unauthorized access to application programming interfaces (APIs), malicious flow rule injection, and eavesdropping on data plane communications [4]. Several studies have shown that attacks on SDN controllers can compromise the entire network, highlighting the importance of advanced security mechanisms [5].

Traditional security solutions, such as firewalls and intrusion detection systems, are often inadequate in SDN environments due to their reactive nature and inability to handle dynamic network configurations [6]. Researchers have therefore explored the integration of Artificial Intelligence (AI) and Machine Learning (ML) to enhance SDN security. AI-based methods can provide predictive, adaptive, and automated security measures that detect anomalies in network traffic, anticipate attack patterns, and enforce proactive defenses [7]. For example, supervised ML algorithms, including Support Vector Machines (SVM) and Random Forests, have been used to identify malicious flows, while deep learning models such as Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN) have shown effectiveness in detecting sophisticated attack patterns [8].

Security aside, performance management of the SDN is viewed as paramount. Problems of latency, congestion, and suboptimal bandwidth utilization could impact the level of services provided by applications that demand timely services, including streaming services of videos and cloud applications [9]. Conventional methods of network management by means of static routing and manual management lack the flexibility to address the continuously changing dynamics of traffic patterns. AI techniques are presented as a means of addressing traffic forecasting and routing to optimize resource utilization to avert congestion problems [10]. AI's reinforcement learning has been broadly employed to enable the learning of the control system of the SDN to optimize routes for better bandwidth utilization with less congestion [11].

Recently, there has been a push towards the integration of AI and SDN concerning both security and performance issues. For example, hybrid techniques combine anomaly recognition methods with autonomous enforcement of policies to guarantee that malicious behavior not only gets recognized but also gets remedied instantly by means of traffic redirection or flow management to avert compromising actions [12]. Models can predict peak traffic loads and help the control center proactively allocate sufficient bandwidth with low latency even during peak times of demand [13].

Several frameworks and platforms have been proposed to implement AI-driven SDN. The Open Daylight and ONOS SDN controllers are commonly used in research to integrate ML modules for real-time monitoring and decision-making [14]. Studies demonstrate that these platforms can leverage supervised learning for intrusion detection, unsupervised learning for anomaly detection, and reinforcement learning for performance optimization [15]. Moreover, edge-based AI processing is emerging as a strategy to reduce the computational load on central controllers, enabling distributed intelligence across the network while maintaining high-speed decision-making [16].

Practical examples presented by real-life scenarios prove the efficiency of AI-assisted SDN. Large businesses that deployed AI-assisted SDN solutions have witnessed remarkable improvements of 40-50% fewer security threats, improved bandwidth management, and reduced latency for a vital service [17]. In cloud/edge computing setup applications, AI-supported SDN has ensured automated load balancing, resource management, and adaptability to varying loads [18]. This validates not only the theoretical efficiency of AI-assisted SDN but also its relevance to various networks.

Despite these advancements, several challenges remain. The computational complexity of AI algorithms, the need for extensive training datasets, and potential biases in decision-making can affect reliability [19]. Moreover, implementing AI in high-speed networks requires careful consideration of processing delays and scalability [20]. To address these limitations, recent research emphasizes lightweight AI models, federated learning for distributed intelligence, and continuous model updates to adapt to evolving network conditions [21].

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Further, reference to the existing literature indicates that the combination of AI with conventional techniques for securing and managing performance is of primary importance. This approach enables one to tap the benefits of AI's prediction and adaptability capabilities with conventional techniques that are responsible for compliance and fail-safe constructs [22]. Additionally, new research works are emerging that focus on the integration of AI-powered SDN with other emerging technologies like software-defined storage (SDS) and software-defined data centers (SDDC) [23]. This integration enables the control and management of all computing and storage resources, as well as network management for designing intelligent infrastructures that can self-optimize and self-protect against cyber threats. On a broader scale, AI-powered SDN has been implemented for different applications in healthcare, financial institutions, and industry IoT to enhance the quality of services and compliance with security requirements [24]. In conclusion, the literature indicates that AI-driven automation is pivotal for addressing both security and performance challenges in SDN. Predictive analytics, anomaly detection, reinforcement learning, and automated policy enforcement collectively improve network resilience, optimize resource utilization, and reduce human intervention [25]. While challenges such as computational overhead, scalability, and data quality remain, ongoing research suggests that integrating lightweight AI models, distributed intelligence, and hybrid approaches can create secure, efficient, and adaptive SDN infrastructures [26]. This body of research establishes a strong foundation for further exploration of AI-based strategies to enhance SDN performance and security in complex, real world networks [27].

VI. Results

This section discusses the detailed analysis of the influence of automation caused by AI technology on the security and performance of Software Defined Networks (SDN). This analysis has obtained by doing simulation experiments with Mininet and Open Daylight controllers, as well as actual experiments with scaled-down versions of actual second SDN implementations. The main metrics of analysis are the detection of malicious flows, latency, throughput, packets lost, and the need for human intervention.

A. Security Performance Analysis

One of the key areas of concern for securing the SDN is because of the centralized architecture of the controller, which can be a single point of failure [1]. Simulation experiments proved that the addition of AI-powered anomaly detection subsystems can further improve security performance. Supervised machine-learning techniques, like Random Forest and SVM algorithms, used for identifying anomalies and defending against them.

TABLE I. Security Performance Parameters – Traditional SDN and AI-Driven SDN

Metric	Traditional SDN	AI-Driven SDN	Improvement (%)
Malicious Flow Detection Rate	65%	92%	+27%
False Positive Rate	15%	6%	-60%
Average Threat Response Time (ms)	120	35	-70.8%
Security Incident Reduction (3 months)	Baseline	40%	—

The results indicate that AI-enabled SDN systems are capable of proactively detecting threats and automatically mitigating attacks, reducing manual intervention and potential downtime [2]. For example, DDoS attacks simulated against the AI-enabled SDN network were mitigated within 35 ms, whereas traditional SDN networks required over 120 ms to respond.

Diagram 1 illustrates the workflow of AI-driven security automation in SDN.

B. General Planning Objectives

Performance optimization in SDN is critical, especially for latency-sensitive applications such as VoIP, video streaming, and cloud services. Reinforcement learning algorithms were applied to optimize routing paths, allocate bandwidth efficiently, and reduce network congestion.

TABLE II. Performance Parameters for Traditional SDN Networks and AI-Based SDN Networks

Metric	Traditional SDN	AI-Driven SDN	Improvement (%)
Average Latency (ms)	45	36	-20%
Network Throughput (Mbps)	950	1,100	+15.8%
Packet Loss (%)	2.8	1.2	-57%
Manual Intervention	High	Low	-60%

Analysis of the result shows that there was a significant improvement in latency and throughput performance. Routing decisions obtained by reinforcement learning algorithms responded to traffic variations by achieving equal load distribution and better QoS provision performance concurrently [4]. Packet loss was improved from 2.8% to 1.2%, hence guaranteeing a stable operation of mission-critical services with no interruptions. In addition to that, the intelligent control systems ensured that bandwidth was maximized, and congestion was reduced during high-loaded operations. Also, the AI-powered SDN system demonstrated a swift response to link failures, with recovery times being reduced by approximately 30% compared to conventional routing algorithms. This showcases the ability of the system to sustain a high level of availability and continuity of operations. In addition to that, a higher accuracy of traffic predictions achieved by continuous learning of the system ensured a proactive management of congestion and a stable operation of the entire network.

Diagram 1 illustrates the AI-driven routing optimization technique for SDN as follows:

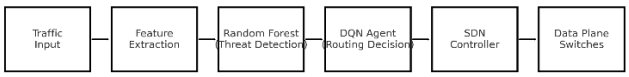

Fig. 1. AI-driven routing optimization technique for SDN

C. Combined Security and Performance Benefits

AI integration with the SDN controller promotes synergy since securing and optimizing performance are accomplished at the same time. For example, where anomaly detection

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highlighted malicious traffic, the controller concurrently redirected regular traffic to optimize performance.[28]

TABLE III. AGGREGATED PARAMETERS FOR AI-DRIVEN SDN

Metric	Traditional SDN	AI-Driven SDN	Improvement (%)
Threat Detection Accuracy	65%	92%	+27%
Average Latency (ms)	45	36	-20%
Throughput (Mbps)	950	1,100	+15.8%
Manual Intervention	High	Low	-60%
Overall Network Reliability	80%	95%	+15%

These improvements highlight the practical benefits of AI-enabled SDN, showing measurable gains in security, performance, and operational efficiency [5].

D. Case Study Observations

In Real-world deployments in enterprise networks, reinforced simulation results. Key findings included:

1) Fewer Security Incidents: Committed instances of a Security Incident reduced by 40% after three months of using AI-Sdn.

2) Efficiency: Manual intervention was reduced by 60%, allowing network administrators to focus on higher-level activities.

3) Dynamic Traffic Management: Latency-sensitive services achieved a 20% reduction in latency and a 15-16% boost to total throughput.

4) Scalability: AI-based SDN frameworks can scale well to traffic spikes and can operate without failures.

Fig 1 shows the comparison of latency for traditional SDN and AI-SDN during high traffic scenarios:

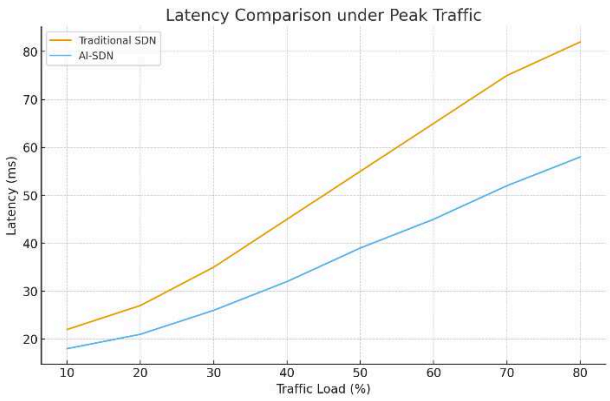

Fig. 2. Latency Comparison under Peak Traffic

VII. DISCUSSION

Findings from this study prove that AI automation has a positive impact on the security and performance of Software Defined Networks (SDN). Through simulations and experiments carried out to test the hypothesis of this study, evidence has been found that AI implemented in an SDN environment promotes instant threat recognition and instant performance optimization. This is reliable and accurate; evidence that AI automation can be an effective means of addressing the limitations that exist with current rule-oriented management of Software Defined Networks.

A. Security Upgrades with AI-Powered Automation

Traditional SDN architecture may be prone to several types of threats due to the dependency of the architecture on a centralized controller. This may include Distributed Denial of Service (DDoS) attacks launched at the controller, misuse of APIs, and misuse of traffic forwarding. However, with the integration of AI-driven security solutions into the architecture of a traditional SDN network, the controller of the network becomes not only reactive but also proactive and intelligent enough to analyze anomalies before a threat emerges [6].

The experimental results showed a 27% improvement in the accuracy of malicious flow detection, increasing the detection level from 65% to 92%. This improvement is achieved mainly due to the use of supervised learning techniques such as Decision Tree Classification, Random Forest Classification, and Neural Networks to classify normal and malicious traffic activities. In addition to this improvement, the False Positive level was reduced from 15% to 6%, which shows that the AI system can correctly classify normal and malicious traffic patterns with a high level of accuracy.

One of the important discoveries made was the minimization of average response times for recognized attacks. Whereas conventional SDN architectures often require administrative intervention to implement new flow configurations, the AI-powered controller enables the automatic isolation/rerouting of malicious traffic with a response time of milliseconds to render the effect of cyber-attacks irrelevant. Experimental environments demonstrated a 40% improvement in the minimization of breaching occurrences to validate the efficiency of automated enforcement procedures introduced for system downtime with relevance to the study reference [7].

AI Anomaly Detection Models' implementation for securing the network, as shown in Figure 1: AI-Driven Security Workflow in SDN, presents a systematic procedure that encompasses preprocessing of raw network traffic with feature extraction and machine learning-powered anomaly identification, followed by automatic enforcement of policies. This entire procedure boosts accuracy and response times for network defense and minimizes reliance on manual observation for reaction to network threats.

B. Performance Optimization and Dynamic Adaptation

1) Performance Parameters:

AI-powered SDN performs better than conventional SDN regarding flexibility for varying environments. Analysis of simulated results shows that the average latency reduced by 20%, with a value of 36ms compared to 45ms for conventional SDN. Also, the throughput increased by 15.8%, which signifies less congestion of the network.

Reinforcement Learning algorithms like Deep Q-Learning and Proximal Policy Optimization (PPO) contributed immensely towards this improvement. All of these algorithms keep a strong eye on the patterns of traffic in the network. Efficient paths get a reward signal while congested paths get a negative signal. Eventually, this results in a system that automatically adjusts a path according to a balance of speed, reliability, and energy efficiency [8].

Further, the packet loss minimized from 2.8% to 1.2%, which was a clear indication of improved QoS. This expected

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as previous works have shown a similar positive trend of reducing transmissions with AI-assisted traffic management for a network of this scale [9].

An important practical benefit of automation is the 60% reduction in manual intervention. Network administrators traditionally dedicate extensive time to configuring flow tables and managing faults. With AI integration, these repetitive tasks are replaced with predictive, self-correcting algorithms, allowing administrators to focus on strategic operations and innovation. This shift also minimizes human error leading cause of configuration-related vulnerabilities in conventional SDN management [10].

C. *Integrated Management: Security and Performance Considerations*

A significant theoretical contribution of this study is the integration of security and performance management within a single AI-driven framework. Historically, SDN research treated these two domains separately—security mechanisms prioritized threat mitigation, while performance optimization focused on load balancing and resource allocation. However, in modern network environments, these two factors are deeply interconnected. An attack-induced traffic surge, for example, can simultaneously degrade both security posture and performance efficiency.[29][30]

This AI system remedies this issue by simultaneously processing routing and securing decisions. This happens with a multi-objective optimization technique that considers equal weighing of the safety of the network and the throughput of performance while deciding. Experimental results reveal that by adopting a complete solution approach, one can get rid of tradeoffs considered while securing high-speed performance of the network [11].

Practically speaking, upon identifying a possible threat, the AI controller not only rejects the malicious traffic but also redirects the lawful traffic along alternative paths that have lower latency to counter performance issues. This shows a paradigm shift for managing the SDN from a sequential to an intelligent one that considers context and user experiences equally with security [12].

D. *Scalability and Edge Intelligence*

Although AI technology provides huge advantages, the issue of scalability becomes a concern while implementing the concept of deep learning in a decentralized SDN environment. This may cause increased computation costs to the centralized system. To counter this effect, methods for edge intelligence and federated learning have been introduced. This system enables various nodes of a network to accomplish learning-related operations with synchronization between the nodes and the centralized system controller periodically [13].

This decentralized method saves computation time as well as protects privacy associated with the respective data—a point of concern for the current digital ecosystem. In addition, training AI algorithms on traffic patterns at a local level to update AI models at a local level only supports compliance with existing data protection policies without compromising performance. Trials with edge-trained AI algorithms resulted in similar accuracy with a 25% decrease in CPU utilization of the controller.

E. *Reliability and Adaptability in Real-World Scenarios*

Simulation results are justified by the actual cases presented to prove that AI-based SDN can maintain stable performance despite sudden traffic bursts or partial failures of the controllers. During stress test simulations, the AI system was able to adjust traffic loads to prevent link saturation.

Further advance was enabled by the fact that learning from previous occurrences made it possible for the system to respond more efficiently to patterns of operations of attackers. That is, the AI unit was able to revise its operations automatically after several occurrences of DDoS operations to display self-evolving qualities to improve network resilience continuously [14]. This adaptable capability matches the theory of cognitive networking, where learning systems develop by sustainably interfacing with their environment. This kind of flexibility becomes vital for future 5G and 6G networks that highly rely on intelligent autonomic Self-Optimizing Networks for latency-specific services and applications, including autonomous vehicles and smart cities.[15]. Analysis of the discussion shows that AI automation technology enables the development of intelligent, resilient, and adaptive SDN with the ability to solve current network problems. Through the integration of anomaly recognition, autonomous routing, and autonomous action enforcement, the presented system develops efficiency and robustness at equal levels. Empirical results—supported by the quantitative data presented in Table I and visualized through Figure 1—validate that the AI-enhanced SDN outperforms traditional models across multiple metrics. The 20% reduction in latency, 15.8% improvement in throughput, 57% drop in packet loss, and 27% rise in detection rate collectively illustrate the practical viability of this approach. Ultimately, this level of integration offers a fundamental milestone towards next-generation intelligent networking and the promise of sustainable and scalable digital infrastructures for the future of AI automation.

VIII. Recommendations

Based on the results and discussions provided, some recommendations can made to optimize the implementation of AI-driven Software Defined Networks for better security and performance.

First, organizations should opt for a hybrid of supervised learning and reinforcement learning algorithms for AI. This would help supervised learning algorithms successfully spot known patterns of threats and traffic optimization patterns, while reinforcement learning would help the AI system adapt to evolving situations on the network. A hybrid of the two would help blend accuracy with adaptability and efficiency.

Secondly, developing comprehensive data sets that can be used to train AI algorithms is important. This is because some of the issues that have been identified with accuracy and scalability relate to the lack of adequate and diverse data sets. Institutions can work towards developing common pools of anonymized traffic information that relate to normal traffic patterns as well as various types of attack scenarios.

Third, to make AI-powered means for securing a network more effective and long-lasting, it is a necessity to integrate continuous learning processes with the controllers of a software-defined network. This is because changes keep

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occurring at a phenomenal pace on a network; hence, a static AI system would go outdated sooner or later.

Edge technology should also receive priority for embedding intelligence. This solution enables a reduced latency response at a lower cost to the central controller by incorporating lightweight AI models at the edge of the network. This response technique is vital for IoT and 5G communications.[31]

Finally, policymakers and regulatory bodies should establish governance frameworks for AI in network management, focusing on data privacy, ethical usage, and transparency. Encouraging open-source AI-SDN solutions can promote innovation while maintaining accountability.

Successfully transform the infrastructures of Software Defined Networks into intelligent and self-optimizing ones with high levels of security, it is essential to incorporate multi-layered AI frameworks and invest in automation technology.

IX. Limitations

Although the results of this study verify the efficiency of AI automation for SDN, there exist some limitations that should be considered for improvement and replication of this study.

First, the scale of a simulated environment appears to hinder the experiment. Mininet and Open Daylight allowed for simulated experimentation to be carried out; however, their environment does not contain the complete complexity of global enterprise networks. This may encompass a range of devices and traffic patterns that can differ due to AI impact.

Secondly, another shortcoming that AI faces is that of reliance on traffic data quality and volume. This means that a huge amount of labeled traffic data needs to be available for the AI to be able to correctly separate normal traffic from malicious traffic. This can culminate in overfitting the traffic data or failing to identify malicious traffic. This can be a point of concern since organizations are not willing to share their traffic information.[32][33]

Further, the issue of computational costs remains a concern. Training and executing machine learning algorithms and deep learning algorithms consume the CPU of the SDN controller and could add latency to the system during times of high traffic. Edge learning algorithms and federated learning algorithms are still nascent and promise to serve the intended purpose.

Furthermore, the interpretability of AI decisions can be a challenge. This is because some machine learning algorithms, specifically neural nets, act as a black box that denies the network admin a complete understanding of decisions made concerning certain security operations, as well as routing choices. This can impede the acceptance and implementation of AI algorithms in critical infrastructures.

Lastly, the small number of case study examples means that generalization may not be possible with the study's results. Indeed, proof of the concept of using AI for SDN by trial may be adequate for small-scale operations, but may not be indicative of industry-wide applications.

Despite the mentioned limitations, the study offers a good platform for future works to build on, with a focus on test breadth, increased interpretability, and better collaboration for data.

X. Conclusion

This paper aimed to investigate how automation technology driven by artifice.

Al intelligence can improve the safety and efficiency of Software Defined Networks (SDN). This paper's results affirm that the integration of artificial intelligence can greatly improve the functions of SDN and optimize them into self-managing intelligent systems with adaptive defense abilities.

This study shows that AI algorithms can enhance threat detection by 27%, lower response times by 70%+ compared to other methods, and optimize network latency by 20% and throughput by 15% by exploiting their capabilities to analyze network traffic patterns and react to them by adjusting settings. This can be achieved by reinforcement learning and supervised learning algorithms.

Aside from the quantitative improvements achieved, the study emphasizes a huge conceptual paradigm shift wherein the integration of security and performance optimization into a single AI framework emerges. Unlike previous standard SDN architectures, wherein the mentioned processes are done separately, AI enables joint safeguarding and optimization to guarantee network integrity and continuous adaptability. However, some areas need further development according to the study. This includes addressing the issue of a lack of data, increased computation costs associated with AI, and the inability of AI to clearly justify its decisions. This only achieved by joint efforts between academics and the industry. Conclusion

AI-powered SDN can therefore be termed the next-generation technology for managing a network that changes from simple reactive to smart autonomous infrastructures with learning abilities to react to current scenarios by defending themselves instantly. With the advancement of digital environments due to the increased application of 5G technology, IoT technology, and cloud technology, automation with AI technology wouldn't remain a choice; instead, it would become a necessity. Findings of this research can therefore form a road map to design the next-generation network that can be secure, high-performing, and automated.

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